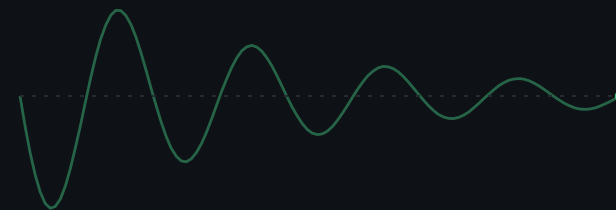


zicato

a self-improving harness for any system you can measure –
evolution under a **protected incumbent**.

a sibling of goldfive + harmonograf



loss settling, generation over generation

Agentic systems drift.

As prompts, tools, models and tasks change, quality silently regresses – and you usually can't tell whether a change helped or hurt.



Measuring real improvement without ground truth.

Four pressures fight every attempt to tell a good change from a bad one.

01

NOISY

LLM stochasticity.

One duel misleads – the same pair re-run disagrees.
A single win can be luck.

02

EXPENSIVE

A board run costs budget.

Every evaluation burns tokens and wall-clock. You cannot re-measure for free.

03

A PROXY

Goodhart's law.

The metric is a stand-in. Push it hard and you game it – you overfit the board.

04

NEVER REGRESS

Protect what worked.

A change must never silently break a task that passed before. Asymmetric risk.

The crux: get a signal that is real, cheap enough, hard to game, and regression-guarded.

A champion / challenger loop.

Mutate the harness into a challenger, race it against the reigning champion, promote on a gate.



NESTED COUNTERS

epochs

one frozen evaluation contract

rounds

one propose / duel / gate

generations

the field a round mints

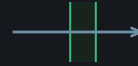
generations nest in rounds nest in epochs

zicato composes proven selection theory.

Each borrowed idea is a small, well-understood mechanism. The loop is their composition.

GATING / MONOTONIC IMPROVEMENT

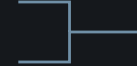
evaluator match + trust regions



AlphaZero-style: promote only on a beat; RL never steps past a region gate

TOURNAMENTS & SUCCESSIVE HALVING

Swiss / elimination / racing



Hyperband from AutoML & bandits – spend evaluation where it counts race

REPLICATION = THE NOISE LEVER

best-arm identification



re-run the duel to shrink variance; more samples, not more brackets replicate

INCUMBENT PROTECTION = ELITISM

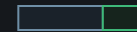
evolution strategies



the champion always survives unless genuinely beaten – elitist carry elitism

ADAPTIVE-DATA-ANALYSIS DEFENCES

the reusable holdout / the Ladder



Dwork et al. & Blum-Hardt: query a fixed set without using it up holdout

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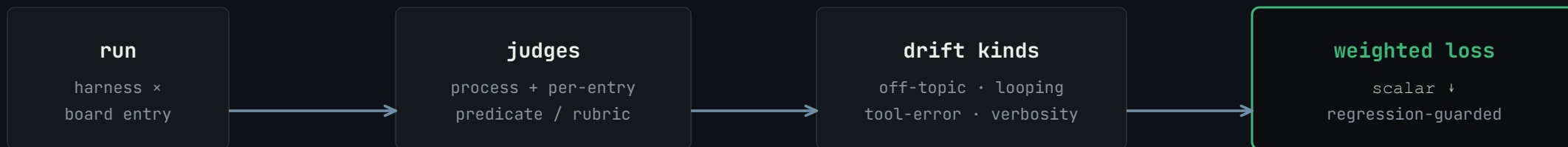
the composition



gate + tournament + replication + elitism + holdout, made to bite on agent behaviour

goldfive's judges turn behaviour into a shaped loss.

Most loops have only coarse pass / fail. zicato runs the inner harness on goldfive – whose custom judges measure **drift**.



THE WEIGHTED SCALAR

```
scalar = drift_weight · drift_loss + pass_weight · (1 - pass_rate)
```

drift_weight	weight on cleanness
pass_weight	weight on correctness
per_kind_weights	elevate / demote a drift kind
per_judge_weights	one step finer – by judge_name

YOU DEFINE WHAT “GOOD” MEANS

in-run process judges + per-entry
predicates and rubrics fold into one
rich, shaped, regression-guarded signal –
where other loops have only thumbs up/down.

This signal is what makes the loop work.

Promotion needs three things at once.

On reject, the champion stands. Asymmetric by design – the loop reduces drift without breaking what worked.

1 scalar margin

loss drops by $\geq \text{promote_margin}$

2 pass-rate monotonicity

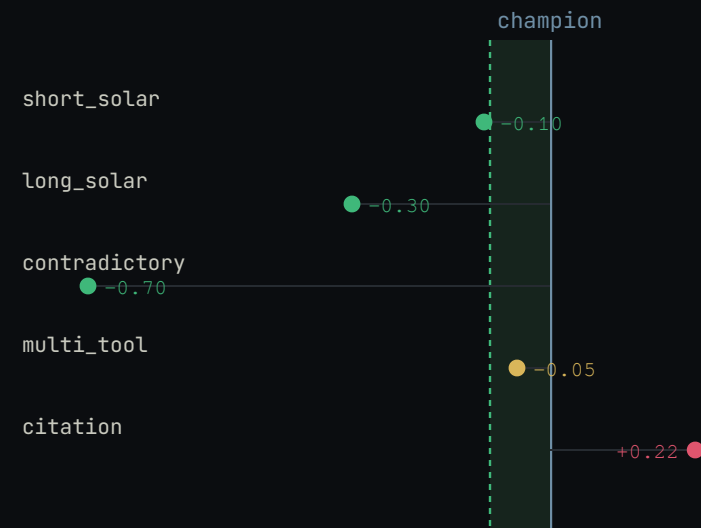
no previously-passing entry regresses

3 no per-task regression

guarded namespaces may not worsen

promote $\Leftrightarrow$ all three hold. else: reject, champion stands.

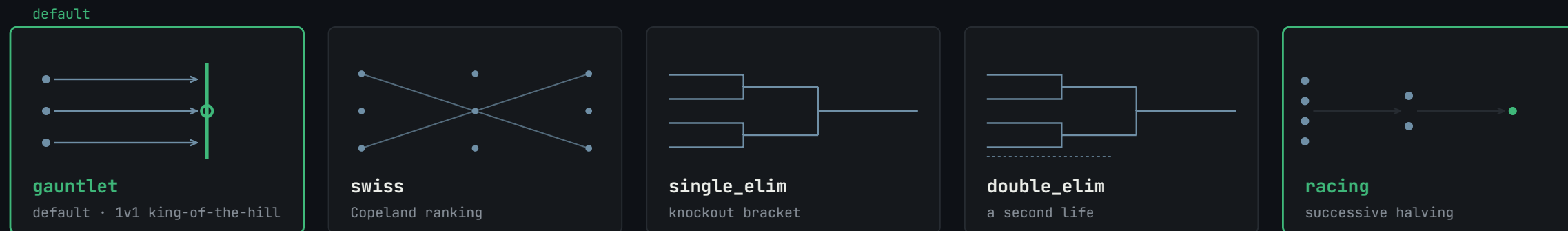
Δ vs CHAMPION · per-entry loss



← improvement promote_margin regression →
 green clears the margin · amber too small · red regresses → reject

Configurable per epoch – but the gate is invariant.

Replication, not bracket shape, is the noise lever. The gate, not the bracket, protects the incumbent.



The proposer is part of the evaluation contract.

Each round it reads the brief, mutation manifest, and loss patterns, and emits one **Experiment**.

(a) SKILL-COMPOSED DEFAULT

drop in skills/*.md – no code

- ☐ grounding.md
- ☐ house-style.md
- ☐ checklist.md

injected into the system prompt;
steers HOW the default proposer reasons.
runs on the auxiliary model.

(b) CUSTOM ADK AGENT

drop in agent.py – its own model

agent.py

tool-using • own model

- read mutation
- propose patch
- reason

its presence selects the tier-(b) path;
its contents are part of the contract.
you own the model + the tools.

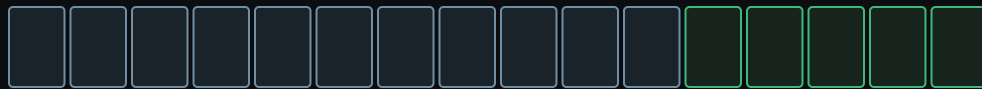
CONTRACT

Changing the proposer or its skills **rolls a fresh epoch** – just like editing the board.
generations proposed under different proposers are not directly comparable.

Overfitting defences – default-on.

The proposer adaptively queries a fixed board, so it can **memorize** it (Goodhart / adaptive data analysis).

TRAIN / HOLDOUT SPLIT

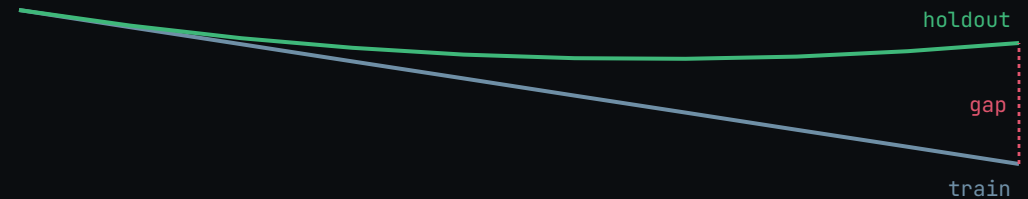


train slice

holdout

proposer + detectors see train only; promotion confirmed on holdout.

GENERALIZATION GAP



gap widens → fake progress → end the epoch

holdout-gated promotion

Ladder / Thresholdout,
budgeted

generalization-gap detector

a zicato health
check

holdout rotation

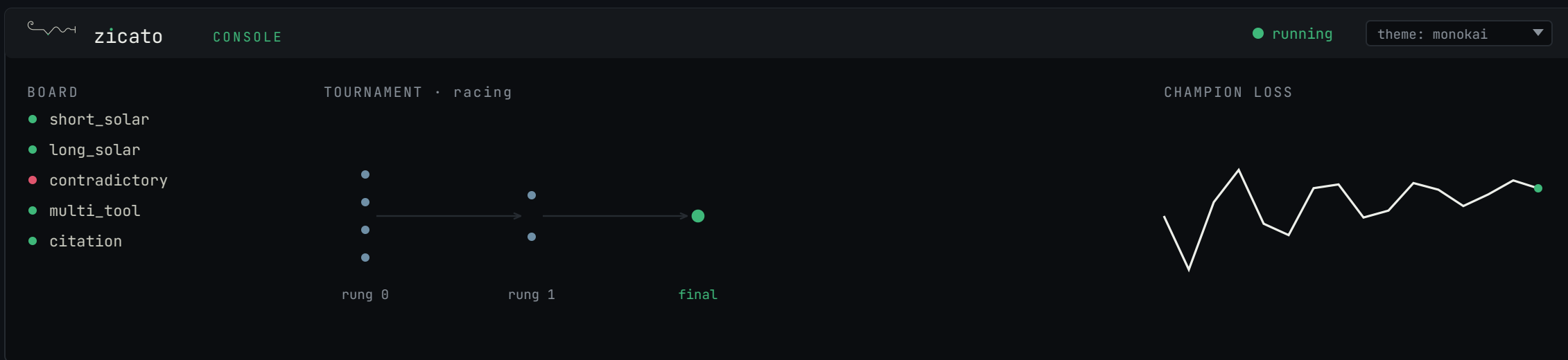
no slice mined
forever

proposer leakage restriction

patterns aggregated, inputs
withheld

Console – a dense observatory.

Live per-structure figures and board status, over a Rust watchdog and JSONL telemetry.



14 themes

monokai default + 13 Gogh palettes, swatch dropdown

tournament builder

a chat copilot edits the draft contract

configurable models

endpoints per harness / auxiliary / proposer



The thesis, in one line.

**Known-good selection theory, made to bite on agent behaviour
via goldfive's drift judges –**
promote only what genuinely, durably beats the incumbent.

WHAT'S NEXT

- Ranked Pairs
 - Bradley-Terry
 - maximal lotteries
- replicate-first, resolve-second.

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